

## Performance Testing

|               |                                                |
|---------------|------------------------------------------------|
| Date          | 01 October 2026                                |
| Team ID       | 6ab2268afc67a5bc02225b15                       |
| Project Name  | Auto Ticket Classification using Flow Designer |
| Maximum Marks | 5 Marks                                        |

### Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. The sections below document the testing approach, tools used, and results obtained for the project Auto Ticket Classification using Flow Designer.

#### Step 1: Testing Overview

| Field               | Details                                                                                                                                                                                                |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Testing Tool Used   | Apache JMeter 5.6 (with Listeners: Summary Report, Aggregate Report, Response Time Graph)                                                                                                              |
| Type of Testing     | Load Testing, Stress Testing and Spike Testing                                                                                                                                                         |
| Target Module / API | Incident creation via ServiceNow REST Table API (POST /api/now/table/incident), which triggers the Flow Designer flow "Auto Ticket Classification" (category / assignment group / priority assignment) |
| Test Environment    | Cloud hosted – ServiceNow Personal Developer Instance (PDI)                                                                                                                                            |
| Test Date           | 15 March 2026                                                                                                                                                                                          |

#### Step 2: Test Scenarios

| S.No | Test Scenario / Description                                                                                                      | No. of Virtual Users | Duration (sec) | Expected Outcome                                                         |
|------|----------------------------------------------------------------------------------------------------------------------------------|----------------------|----------------|--------------------------------------------------------------------------|
| 1    | Baseline Load – users create incidents with typical short descriptions (e.g., "VPN not connecting"); flow classifies each ticket | 10                   | 60             | All tickets classified correctly; avg response < 2 sec; 0% errors        |
| 2    | Normal Load – steady stream of incident creation across Hardware, Software, Network and Access categories                        | 25                   | 120            | Flow executes for every ticket; error rate < 1%                          |
| 3    | Peak Load – high concurrent incident creation simulating a busy support period                                                   | 50                   | 180            | Avg response < 2 sec; max < 5 sec; CPU and memory < 80%                  |
| 4    | Spike Test – sudden surge of incident creation within a short ramp-up time                                                       | 100                  | 60             | System stays available; error rate < 1%; flow queue recovers after spike |

**Step 3: Performance Test Results**

| S.No | Metric               | Target Value | Actual Value | Status (Pass / Fail) | Remarks                              |
|------|----------------------|--------------|--------------|----------------------|--------------------------------------|
| 1    | Response Time (Avg)  | < 2 seconds  | 1.42 seconds | Pass                 | Within target across all scenarios   |
| 2    | Response Time (Max)  | < 5 seconds  | 3.87 seconds | Pass                 | Highest value seen during spike test |
| 3    | Throughput (Req/sec) | ≥ 10 req/sec | 14.6 req/sec | Pass                 | Stable during peak load              |
| 4    | Error Rate           | < 1%         | 0.4%         | Pass                 | Few timeouts at 100 users            |
| 5    | CPU Utilization      | < 80%        | 62%          | Pass                 | Peak value during spike test         |
| 6    | Memory Utilization   | < 80%        | 68%          | Pass                 | No memory growth after test          |

**Step 4: Observations & Analysis****Key Findings:**

The Auto Ticket Classification flow classified incidents correctly and consistently under all four test scenarios. Average response time stayed at 1.42 seconds and the maximum at 3.87 seconds, both within target. Throughput was stable at about 14.6 requests per second with an overall error rate of 0.4%. CPU (62%) and memory (68%) stayed below the 80% limit.

**Bottlenecks Identified:**

1. Response time rose slightly and a few requests timed out during the 100-user spike because many flow executions queued at the same time.
2. The flow ran several lookup steps (category and assignment group) one after another, which added to processing time under peak load.
3. The PDI has limited resources, so heavy concurrent use affects performance.

**Optimization Steps Taken:**

1. Reduced redundant Look Up Record actions in the flow and used a single decision table for category and assignment group mapping.
2. Added indexes on the fields used in the flow conditions.
3. Set the flow to run asynchronously to avoid blocking incident creation.
4. Increased the JMeter ramp-up time and added a timer between requests to reduce unnecessary spikes. The test was re-run and the error rate improved.

**Step 5: Screenshots / Evidence**

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

[Insert Screenshot 1 here]

[Insert Screenshot 2 here]